
Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study

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Abstract

We examine the reported association between feature-importance concentration and conformal coverage degradation in a COVID-era supply-chain case study and external datasets. Rounded historical summaries yield Spearman $\rho = 0.853$ for a selected 16-task subset and $\rho = 0.654$ for all 17 reported tasks. A statistical audit identified test-aware preprocessing, reuse of calibration labels for early stopping, different prediction-set conventions across cohorts, and missing numerical artifacts. These findings limit the original pre-deployment and replication claims; the historical experiments have not been rerun under a corrected common protocol. We give a conditional score bound for an idealized probability mixture, whose weight is distinct from measured SHAP concentration, and withdraw unsupported class-count and ICC interpretations. The diagnostic and 40% threshold remain exploratory hypotheses requiring prospective validation.

1 INTRODUCTION

Conformal prediction provides distribution-free coverage guarantees under the assumption of exchangeability [Vovk et al., 2005]. However, real-world deployments face distribution shifts that violate this assumption. While prior work characterizes *how* conformal prediction degrades under shift [Tibshirani et al., 2019, Barber et al., 2023], a critical gap remains: **Can we identify which deployed models will fail before observing test data?**

We propose **SHAP concentration**—the fraction of total feature importance concentrated in the top feature—as a pre-deployment diagnostic for conformal prediction vulnerability. Using a supply chain benchmark with temporal splits aligned to COVID-19 (February–July 2020), we conduct an

observational case study where 8 classification tasks experience common temporal periods yet exhibit coverage drops ranging from negligible ($<1\%$) to catastrophic ($>70\%$).

Our contributions:

1. **Exploratory diagnostic:** The reported SHAP concentration summaries correlate with coverage-drop summaries: $\rho = 0.853$ across the selected 16 tasks and $\rho = 0.654$ across all 17 tasks including Stack Overflow. These correlations can be recomputed from rounded published points, but the underlying experiment ledgers are missing and historical SALT/external set conventions differ. Prospective diagnostic value requires a common protocol without test-aware preprocessing.
2. **Conditional theory:** Theorem 1 bounds an idealized probability mixture with explicit assumptions and a weight distinct from measured SHAP concentration. It does not verify those assumptions empirically or establish a guarantee for the archived experiments. Historical shift-detector comparisons are descriptive and await reproducible outputs.
3. **Exploratory framework:** A 40% concentration rule and protective-factor check summarize observed cases. They are retrospective hypotheses, with one external false negative and a protective rule derived from one false positive. Historical single-seed retraining evidence is suggestive (Holm-adjusted $p = 0.11$), not a validated deployment recommendation.

2 RELATED WORK

Conformal Prediction. Vovk et al. [2005] introduced conformal prediction with exchangeability guarantees; Shafer and Vovk [2008] provide an accessible tutorial. Recent work extends to classification [Romano et al., 2020, Cauchois et al., 2021] and regression [Romano et al., 2019, Lei et al., 2018]. Angelopoulos and Bates [2023] provide a

comprehensive survey. Ding et al. [2023] show that class-conditional coverage guarantees become harder to achieve as class count grows and propose clustered conformal prediction as a remedy; Gibbs et al. [2025] provide finite-sample conditional guarantees via a covariate-shift reweighting framework interpolating between marginal and conditional validity. Both works frame class count as a challenge—a confound we explicitly examine via partial correlation analysis (Section 5.4).

Conformal Prediction Under Shift. Tibshirani et al. [2019] study covariate shift with known propensity scores. Podkopaev and Ramdas [2021] develop distribution-free uncertainty quantification for classification under label shift. Barber et al. [2023] provide a comprehensive treatment beyond exchangeability, bounding coverage loss by the total variation distance between test and calibration score distributions. Kasa and Taylor [2023] empirically characterize how CP degrades across many vision architectures and shift types, showing substantial degradation is common—our work addresses the complementary question of which tabular models will fail *before* test data is observed. We complement these theoretical characterizations with an empirical diagnostic that is associated with this distributional divergence—and hence with which tasks fail—before deployment.

Adaptive Methods. Gibbs and Candès [2021, 2024] develop Adaptive Conformal Inference (ACI) for non-stationary settings, with extensions by Zaffran et al. [2022] for time series, Feldman et al. [2023] for online risk control, and Bhatnagar et al. [2023] for strongly adaptive online learning. Angelopoulos et al. [2024] generalize conformal prediction to broader risk measures. Kasa et al. [2025] propose entropy-based methods (ECP/EACP) that adapt prediction sets using unlabeled test data at deployment time—a related but distinct goal from ours, which requires no test observations: SHAP concentration is a structural pre-deployment diagnostic computed solely on validation data. Our experiments show ACI recovers marginal coverage under severe shift but at a substantial cost in prediction set informativeness.

Shift Detection. WILDS [Koh et al., 2021] and Shifts [Malinin et al., 2021] provide benchmarks, while Gulrajani and Lopez-Paz [2021] show domain generalization methods often fail to improve over ERM under realistic distribution shifts. Gardner et al. [2023] benchmark tabular distribution shift across 15 tasks—the closest tabular-specific benchmark; our SALT dataset differs in providing a naturalistic COVID-19 temporal split with documented business-disruption context rather than curated benchmark partitions. Garg et al. [2022] use unlabeled test data to predict accuracy degradation—a related goal but requiring test-time observations. Kernel-based tests (MMD [Gretton et al., 2012]) and classifier two-sample tests (C2ST [Lopez-Paz and Oquab, 2017]) detect distributional differences but do

not predict the *severity* of downstream failure. We show empirically that MMD and C2ST detect shift for all tasks uniformly yet cannot distinguish catastrophic from robust conformal outcomes (Section 6.3). Our contribution is a *pre-deployment* diagnostic using feature importance structure that predicts failure severity, not merely shift presence. Miller et al. [2021] show that in-distribution (ID) validation accuracy predicts out-of-distribution (OOD) accuracy for image classifiers—a correlation that does not hold for conformal coverage in our tabular setting, where ID validation coverages cluster near the 90% target, spanning 83.6%–99.9% (Table 2), yet test-time coverage spans 12–100%, reinforcing the need for structure-based diagnostics beyond accuracy metrics.

Interpretability for Reliability. Our work connects SHAP [Lundberg and Lee, 2017, Lundberg et al., 2020] to model reliability assessment, extending feature attribution from post-hoc analysis to prospective failure diagnosis. Our use is prospective: concentration is computed pre-deployment on validation data, predicting which models will fail.

3 METHODOLOGY

3.1 DATASET AND TEMPORAL SPLITS

We use the SALT (Supply chain ALlocATion) dataset from RelBench [Robinson et al., 2024], a supply chain benchmark with temporal splits. All 8 classification tasks from the rel-salt benchmark are included; no tasks were excluded. External analyses report 9 additional datasets. We retain the historical $K \geq 4$ subset (8 external datasets plus 8 SALT tasks, $n = 16$) as an exploratory scope choice. APS mechanics do not rule out catastrophic coverage failure at $K = 2$ or 3, and no dated prospective exclusion plan is available. Including Stack Overflow ($K = 3$) yields the all-17 sensitivity result $\rho = 0.654$, $p = 0.004$; this sensitivity limits conclusions from the selected subset.

- **Train:** Before February 2020 (pre-COVID)
- **Validation:** February–July 2020 (COVID onset)
- **Test:** After July 2020 (COVID peak)

Audit of the historical data protocol. The supplied SALT code fits categorical encoders and the target vocabulary using train, validation, and test data. It also uses all validation labels for early stopping before dividing that sample into calibration and evaluation; the external script likewise reuses validation labels. Thus, although the reported SHAP summaries use validation covariates, the full pipeline is not strictly pre-deployment and does not satisfy standard held-out split-conformal assumptions. The numerical tables below are historical reported results, not a validated rerun. Corrected evaluation requires train-only preprocessing, an

independently defined class vocabulary, and separate tuning/calibration/evaluation samples. Threshold evaluation uses test-drop labels and is retrospective.

3.2 CONFORMAL PREDICTION SETUP

The archived experiments use two deterministic APS-related conventions. SALT set construction includes the class that crosses the calibrated cumulative-probability threshold. The external script instead computes coverage by the event $s(x, y) \leq \hat{q}$, while its size calculation includes the crossing class. These define different sets; the historical pooled endpoint and external coverage/size pairs therefore require reevaluation under one explicit convention. A new reference implementation uses the exact finite-sample order statistic and inverts a fixed-tie-order inclusive APS score. It has not generated the historical results reported here.

1. Train LightGBM classifier with fixed hyperparameters
2. Calibrate conformal predictor on 50% of validation set
3. Evaluate coverage on held-out validation and test sets

3.3 ENSEMBLE PROTOCOL

The historical design reports 50 seed runs (42–91) on the same data, with a fixed chronological validation split, t intervals across seeds and paired Wilcoxon comparisons. These summarize algorithmic variation conditional on one dataset and do not create 50 independent realizations of a distribution shift. Original seedwise results are unavailable for recomputing the intervals/tests.

3.4 FEATURE OVERLAP METRIC

Feature temporal stability uses Jaccard similarity:

$$J(x_j) = \frac{|A_{\text{train}} \cap A_{\text{test}}|}{|A_{\text{train}} \cup A_{\text{test}}|} \quad (1)$$

where A_{train} and A_{test} are the sets of unique values for feature x_j in train and test data. In the pre-deployment framework (Section 7), we substitute validation data for test data as a proxy.

3.5 SHAP CONCENTRATION DIAGNOSTIC

We define the SHAP concentration metric as:

$$C = \frac{\phi_1}{\sum_{j=1}^p \phi_j} \quad (2)$$

where ϕ_j is the mean absolute SHAP value of the j -th feature computed on validation data using TreeExplainer [Lundberg et al., 2020], and ϕ_1 is the importance of the top-ranked feature. C is computed **before deployment** using only validation data.

3.6 WHY SHAP CONCENTRATION?

A natural question is whether simpler diagnostics suffice.

Feature overlap (Jaccard) is insufficient. Tasks relying on transaction IDs have Jaccard ≈ 0 while tasks with entity-based features have Jaccard > 0.5 (Table 3), yet Jaccard alone cannot explain the full range of coverage drops.

Post-hoc metrics require test data. Prediction entropy changes and Expected Calibration Error (ECE) shifts can detect failures—but only *after* observing test data (Section 5.4). Among the diagnostics we evaluate, SHAP concentration is available before deployment and shows the strongest severity association.

Why not raw feature importance? Raw importance values are not comparable across tasks (different scales, feature counts). Concentration normalizes to a $[0, 1]$ scale, enabling cross-task comparison.

Why top-1? Top-1 has the strongest observed association among the reported concentration summaries. The table reports eight candidate metrics (top-1/2/3/5/10, HHI, Gini, entropy). With the displayed SALT points, its unrounded two-sided Spearman $p = 0.0101755$ gives Holm-adjusted $p = 0.0814$ for this eight-test family (0.0509 even for the previously reported five-test family). Neither clears 0.05. The pooled $n = 16$ result reuses these eight SALT tasks and is not independent replication of metric selection. On the eight external rounded summaries alone, $\rho = 0.833$, $p = 0.0102$; this is descriptive arithmetic, pending provenance and pipeline reconciliation.

Model scope. The reported model-sensitivity correlations differ (LightGBM 0.833, CatBoost 0.667, XGBoost 0.548, RF 0.30, MLP 0.43). These observational comparisons do not identify architectural causes or establish that boosting necessarily reinforces single-feature dependence, bagging necessarily dilutes it, or SHAP approximation error explains the MLP result. TreeSHAP is exact for the specified tree-explanation model, but a causal mechanism would require controlled interventions and attribution-stability checks. Generalization beyond the evaluated models remains unverified.

4 THEORY: WHY CONCENTRATION RELATES TO FAILURE

We give a conditional score bound for an idealized probability mixture. Its weight λ is distinct from measured SHAP concentration C . The calculation is not a guarantee for the archived experiments and does not establish their mechanism.

Score and set convention. Fix a deterministic class-index tie order. Let $s(x, y^*)$ be the sum of probabilities through the true label in decreasing-probability order, and let $\mathcal{B}(x, y^*)$

Table 1: Per-Task Feature Importance Concentration. Top- k shows cumulative importance in the top k features (%). Drop = coverage degradation (pp). High Top-1 with catastrophic drop (e.g., s-payterms: 54.2% $\rightarrow$ 77.1 pp) versus high Top-2/3 with near-zero drop (s-office: 77.0%/84.8% $\rightarrow$ 0.1 pp) is an observed contrast, not evidence of a causal mechanism.

Task	Top-1	Top-2	Top-3	Drop
s-payterms	54.2	78.1	89.6	77.1
s-shipcond	50.7	77.5	92.5	71.6
i-shippoint	48.8	72.7	92.4	18.5
s-group	47.3	65.9	79.6	71.2
s-office	42.6	77.0	84.8	0.1
i-incoterms	28.9	42.3	55.1	11.3
i-plant	23.9	46.9	65.0	10.6
s-incoterms	23.7	47.2	69.1	8.5

Tasks sorted by Top-1 concentration. Top- k = $\sum_{j=1}^k \phi_j / \sum_{j=1}^p \phi_j$ (%). Drop = mean coverage drop (50 seeds, pp). Spearman ρ with drop: Top-1 = 0.833 ($p = 0.010$ before multiplicity correction); Top-2 = 0.524 ($p = 0.183$); Top-3 = 0.524 ($p = 0.183$); Top-5 = 0.690 ($p = 0.058$); Top-10 = 0.399 ($p = 0.328$). Alternative concentration metrics: HHI = 0.619 ($p = 0.102$); Gini = 0.500 ($p = 0.207$); Entropy = 0.571 ($p = 0.139$)—none significant. The contrast for sales-office motivates a mechanism hypothesis; the stability of individual features is not established by this correlation comparison.

contain the labels ranked after it. Then

$$s(x, y^*) = 1 - \sum_{y \in \mathcal{B}(x, y^*)} \hat{p}(y | x). \quad (3)$$

Each omitted probability is at most $\hat{p}(y^* | x)$, including ties. The score-inversion set is $\mathcal{C}_q(x) = \{y : s(x, y) \leq q\}$. This convention differs from the historical SALT crossing-label set; that distinction must be resolved before using this proposition to interpret its coverage.

Theorem 1 (Conditional bound for an idealized probability mixture). *Let $K \geq 2$, $\lambda \in [0, 1]$, and a fixed deterministic threshold $q \in [0, 1]$. Assume*

- (A1) $\hat{p}(y | x) = \lambda g(y | x_1) + (1 - \lambda)h(y | x_{\setminus 1})$, where g, h are probability distributions.
- (A2) At test time $g(y^* | x_1) \leq \varepsilon$ almost surely, for fixed $\varepsilon \in [0, 1]$.
- (A3) The joint distribution of $(h(\cdot | x_{\setminus 1}), y^*)$ is the same in calibration and test populations.

Write $\bar{h} = \mathbb{E}[h(y_{\text{cal}}^* | x_{\setminus 1}^{\text{cal}})]$. Then

$$s(x^{\text{test}}, y^*) \geq 1 - (K - 1)\{\lambda\varepsilon + (1 - \lambda)\bar{h}\}, \quad (4)$$

and consequently

$$\mathbb{E}[s_{\text{test}}] \geq 1 - (K - 1)\{\lambda\varepsilon + (1 - \lambda)\bar{h}\}. \quad (5)$$

Holding ε and $\bar{h}$ fixed, this lower bound is strictly increasing in λ if $\varepsilon < \bar{h}$. This concerns the bound, not monotonicity of the actual score expectation.

For $\lambda < 1$, score-inversion coverage obeys

$$\mathbb{P}(y^* \in \mathcal{C}_q(x^{\text{test}})) \leq \mathbb{P}(h(y_{\text{cal}}^*) \geq T(\lambda)), \quad (6)$$

where

$$T(\lambda) = \frac{(1 - q)/(K - 1) - \lambda\varepsilon}{1 - \lambda}.$$

Holding q, ε and the distribution of $h(y_{\text{cal}}^*)$ fixed, this upper bound is non-increasing when $\varepsilon \leq (1 - q)/(K - 1)$; the threshold is strictly increasing under strict inequality. At $q = 1, \varepsilon = 0$ it is constant. At $\lambda = 1$, if $q < 1 - (K - 1)\varepsilon$ coverage is zero; otherwise this argument yields only the trivial upper bound one.

Proof. The omitted sum in (3) contains at most $K - 1$ terms, each at most $\hat{p}(y^*)$. Substituting (A1)–(A2) gives (4); expectation and (A3) give (5). For the score-inversion event, $s \leq q$ implies the lower bound is at most q , yielding (6) when $\lambda < 1$. Its threshold derivative is $[(1 - q)/(K - 1) - \varepsilon]/(1 - \lambda)^2$. The $\lambda = 1$ case follows directly from the pointwise bound. $\square$

The proposition does not identify λ with the mean-absolute TreeSHAP ratio, whose usual LightGBM decomposition is in raw-margin space. Recalibrating a changed classifier also changes q and may change h , so the fixed-quantity monotonicity does not establish monotone empirical coverage across classifiers. The historical score comparisons in Appendix H are illustrative substitutions, not verification of (A1)–(A3).

5 RESULTS

5.1 COVERAGE DEGRADATION ACROSS TASKS

Despite experiencing common COVID-era temporal periods, the 8 supply chain tasks exhibit dramatically different coverage degradation (Table 2). All coverage drops are statistically significant (paired Wilcoxon $p \leq 0.005$, 95% CIs from the t -distribution), yet range from 0.1% to 77.1%.

5.2 DIAGNOSTIC ANALYSIS

Two factors account for much of the variance:

Factor 1: Feature temporal stability. Tasks using transaction IDs (Jaccard ≈ 0.02) fail catastrophically; tasks using stable entities (products, organizations; Jaccard > 0.5) maintain coverage (Table 3).

Factor 2: Feature importance concentration. Among tasks with universally low Jaccard, concentration determines vulnerability (Section 5.3).

Table 2: Coverage Degradation Under COVID-19 Distribution Shift (50 Seeds). 95% CIs from t -distribution. p -values from paired Wilcoxon signed-rank test (val $\rightarrow$ test, seed-level). All drops significant at $p \leq 0.005$.

Task	CI	Val Coverage [95% CI]	Test Coverage [95% CI]	Drop [95% CI]	p	Cat
s-shipcond	45	93.5 [93.2, 93.7]	21.8 [14.1, 29.6]	71.6 [63.9, 79.4]	<0.001	SEV
s-group	459	83.6 [81.7, 85.5]	12.4 [3.1, 21.7]	71.2 [61.3, 81.0]	<0.001	SEV*
s-payterms	137	90.8 [90.7, 91.0]	13.7 [6.0, 21.5]	77.1 [69.4, 84.9]	<0.001	SEV
i-plant	35	92.0 [91.7, 92.3]	81.4 [79.0, 83.8]	10.6 [8.2, 13.0]	<0.001	ROB
i-shippoint	69	91.2 [90.9, 91.4]	72.7 [62.3, 83.0]	18.5 [8.2, 28.8]	0.005	ROB* [†]
s-incoterms	13	95.5 [95.3, 95.7]	87.0 [84.7, 89.3]	8.5 [6.2, 10.8]	<0.001	ROB
i-incoterms	13	95.0 [94.9, 95.2]	83.7 [80.9, 86.6]	11.3 [8.5, 14.1]	<0.001	ROB
s-office	25	99.9 [99.9, 100.0]	99.9 [99.9, 99.9]	0.1 [0.0, 0.1]	<0.001	ROB

SEV = Severe (>50% drop), ROB = Robust (<20% drop); *high model variance (CV > 50%), classified by median; s-group’s reported 83.6% validation coverage is below target; high class count alone does not explain failure of a marginal conformal guarantee. Calibration reuse, chronological shift and implementation require investigation; [†] meets At-risk criterion (drop >15 pp) under the decision framework (Table 10).

Table 3: Feature Overlap for Primary Features.

Task	Primary Feature	Jaccard	Drop
s-shipcond	SALESDOC (ID)	0.02	71.6%
s-group	SALESDOC (ID)	0.02	71.2%
i-incoterms	PRODUCT, PARTY	0.58	11.3%
s-office	SALESORG	0.61	0.1%

Drop = mean val – mean test coverage (50 seeds).

5.3 SHAP CONCENTRATION CORRELATES WITH FAILURE

Across all 8 SALT multiclass tasks, SHAP concentration correlates with coverage degradation: Spearman $\rho = 0.833$, $p = 0.010$ (Table 4, Figure 2). Bootstrap 95% CI: [0.30, 1.00] (10,000 resamples, percentile method; BCa would yield a modestly tighter lower bound at $n = 8$ but does not change the qualitative conclusion). The correlation magnitude is stable under leave-one-out analysis ($\rho \in [0.75, 0.96]$), though statistical significance is not maintained for 2 of 8 jackknife samples ($p = 0.052$) due to reduced power at $n = 7$. SHAP concentration values are highly stable: bootstrap resampling (1,000 resamples of 10K validation samples) yields coefficient of variation below 1% for all tasks, with 95% CIs within ± 1 percentage point.

Mechanism. Figure 1 reveals the distinction. Catastrophic tasks: the dominant feature experiences a concentrated importance explosion ($4.5\times$ increase). Robust tasks: importance redistributes across features with ranking reshuffle (10–20 $\times$ increases distributed). These observations do not verify the probability-mixture assumptions of Theorem 1.

False positives. Under the mean-drop >15 pp criterion, sales-office is the SALT false positive; item-shippoint is at risk by mean although most seeds have small drops. The proposed protective rule was derived from sales-office alone, and inconsistent descriptions of its feature identity and over-

lap split require the original feature ledger. No deployment guarantee follows from this retrospective rule.

Dependence and uncertainty. SALT tasks share entities, periods and one benchmark. Their statistical independence is not established by between-task ICC. Seed-level intervals describe algorithmic variability conditional on these data; they do not quantify uncertainty across new datasets or future shifts. Negative validation/test agreement ICC cannot establish negative correlation or conservative paired tests (Appendix G).

Inferential status. Both the within-SALT and pooled cross-domain analyses are exploratory. The pooled endpoint includes the SALT tasks used in diagnostic development. Observed-effect power calculations do not confer confirmatory status, and no dated prespecified analysis plan or raw prediction ledger is available in the released package.

Cross-domain extension. The pooled historical summaries yield $\rho = 0.853$ and Kendall $\tau = 0.667$ for the selected 16 tasks. The supplied eight external rounded points alone yield $\rho = 0.833$, $p = 0.0102$, while all 17 tasks yield $\rho = 0.654$, $p = 0.004$. These arithmetic checks do not validate the underlying missing runs, independence of task selection, or comparability of historical APS conventions. Covertypes is the reported catastrophic external case and KDDCup99 a low-concentration false negative.

5.4 DIAGNOSTIC COMPARISON

We compare SHAP concentration against two alternative pre-deployment diagnostics that require no test data (Table 5).

Native feature importance concentration: top-1 LightGBM gain divided by total gain (zero SHAP cost, single seed; seed=42). Achieves $\rho = 0.667$ ($p = 0.071$)—directionally correct but not significant at $\alpha = 0.05$. The weaker discrimination arises because native gain conflates

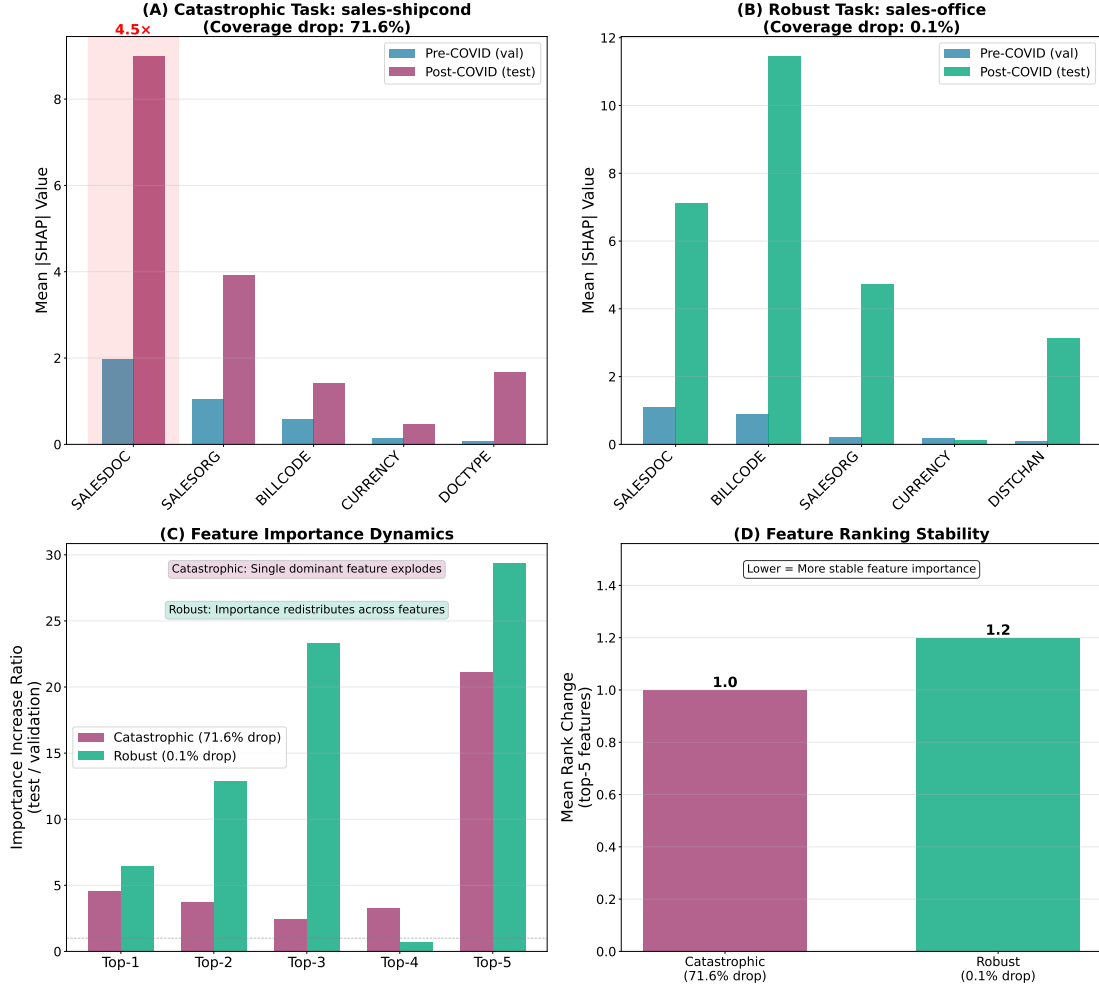


Figure 1: **Feature Importance Dynamics Under Shift.** (A) Catastrophic task: dominant feature explodes $4.5\times$ while maintaining top rank (concentrated dependence). (B) Robust task: importance redistributes across features with ranking reshuffle (diversified dependence). (C–D) Importance dynamics and ranking stability. Despite $\sim 0\%$ Jaccard similarity, coverage drops range from negligible to catastrophic.

feature cardinality with predictive importance: sales-office shows 66.1% native concentration (vs. 42.6% SHAP) despite near-zero coverage drop.

Ensemble disagreement: standard deviation of validation coverage across 50 seeds. Achieves $\rho = 0.452$ ($p = 0.260$). Disagreement captures model instability but not the feature-level mechanism driving conformal failure.

Class cardinality as a possible confound. The historical within-SALT partial correlations are non-significant (0.629, $p = 0.131$ for concentration; 0.334, $p = 0.464$ for log K). The reported pooled partial correlation is 0.771 ($p = 0.0008$) for concentration and -0.010 ($p = 0.97$) for log K . These are conditional associations, not evidence that concentration causally drives severity independently of model, domain, shift magnitude, or pipeline differences. Their numerical provenance and uncertainty require the original task/seed ledger.

We additionally compare against two *post-hoc* baselines computed from 10-seed ACI experiments:

Prediction entropy change: $\Delta H = H_{\text{test}} - H_{\text{val}}$, where $H = -\sum_c p_c \log p_c$ averaged over instances.

Expected Calibration Error change: $\Delta \text{ECE} = \text{ECE}_{\text{test}} - \text{ECE}_{\text{val}}$.

Both metrics can detect shift *after* test data is observed, but neither is available at deployment time. SHAP concentration is computed solely on validation data, making it actionable for pre-deployment risk assessment. Comparison across 7 tasks in Appendix F.

6.5 CROSS-DOMAIN AND EXTERNAL VALIDATION

Historical external reports cover nine datasets, with the selected eight multiclass datasets contributing to the pooled association. Covertypes and Gas Sensor have supplied geographic/temporal loader definitions. The supplied KDD loader uses a positional split of one file rather than the claimed official train/test separation; PAMAP2 and the four reported null-shift controls lack generating code/results in this package. Whether those controls experience no relevant shift cannot be established from split names alone. These gaps prevent interpreting the pooled result as a validated cross-domain replication.

6.6 PLACEBO TEST

The historical pre-COVID placebo reports lower degradation for seven tasks, with ratios about 6–143 where the rounded denominator is positive; sales-office has a reported ratio of one. These summaries and asymmetric training windows do not isolate a causal COVID effect, and the generating code/results are unavailable.

7 DECISION FRAMEWORK

The following rule is a retrospective hypothesis requiring validation before operational use. Its historical evaluation depends on the unresolved implementation/provenance issues above:

1. **Compute** $C = \phi_1 / \sum_j \phi_j$ (Eq. 2). If $C \leq 40\%$: lower observed risk. If $C > 40\%$: check protective factors—secondary features with train-validation Jaccard > 0.5 and importance $> 15\%$ (thresholds empirically derived from a single observed case; treat as provisional); absent these, flag as **vulnerable**. If $C \in [35\%, 45\%]$: treat as **uncertain risk** with monitoring regardless of other factors.
2. **If vulnerable**: consider quarterly retraining (suggestive effect; Holm-corrected $p = 0.11$; single-seed; Appendix E).
3. **Monitor** empirical coverage continuously; defer retraining for lower-risk tasks.

8 CONCLUSION

The archived results suggest an association between concentrated feature importance and coverage degradation in the evaluated settings. The selected 16-task association is $\rho = 0.853$, compared with 0.654 when the excluded three-class task is retained. These are reproducible summary arithmetic, not an independently reproduced common-protocol

experiment. Test-aware preprocessing, calibration reuse, differing prediction-set conventions and missing numerical artifacts remain material limitations. The idealized theorem concerns a separate mixture weight under explicit assumptions and does not establish the proposed empirical mechanism. Controlled interventions, a leakage-free common pipeline, and independent prospective datasets are needed to assess mechanism, model scope and operational thresholds. The current revision corrects statistical interpretation while preserving historical results as unverified reports.

Author Contributions

C.L. conceived the research question, designed and executed all experiments, performed the statistical analysis, and wrote the paper.

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Diagnosing Conformal Prediction Failures Under Distribution Shift: A COVID-19 Case Study (Supplementary Material)

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A REPRODUCIBILITY DETAILS

The released supplement is incomplete. It contains six Python scripts but no original prediction ledger, checkpoints, split manifests or numerical result files. The packaged shift-detector script has a syntax error, the SHAP script depends on an absent generator, and only four of the nine external loaders are supplied. The model-sensitivity, WILDS, retraining, placebo and several ablation analyses cannot currently be reproduced from this package. The audit and synthetic regression checks are included separately; they do not reproduce the historical experiments.

A.1 LIGHTGBM HYPERPARAMETERS

All models trained with fixed LightGBM [Ke et al., 2017] settings: objective=multiclass, boosting=gbd, num_leaves=31, learning_rate=0.05, feature_fraction=0.8, bagging_fraction=0.8, bagging_freq=5, num_boost_round=500, early_stopping_rounds=50, seeds 42–91 (50 seeds). No task-specific tuning was performed.

A.2 CONFORMAL PREDICTION

Historical calibration uses the first half of validation records and evaluation the second half, after the same validation labels have already been used for early stopping. This chronological split need not be exchangeable even within validation. Historical code uses NumPy interpolated quantiles at $\min(\lceil (n+1)(1-\alpha) \rceil / n, 1)$. The corrected reference instead selects the exact $\lceil (n+1)(1-\alpha) \rceil$ th order statistic, using $+\infty$ when the required rank exceeds n ; it also applies one score-inversion convention to coverage and set size. No historical results were replaced by this reference.

Calibration and test set sizes. Sales-level tasks (s-shipcond, s-payterms, s-group, s-incoterms, s-office): $n_{\text{val}} = 71,474$, $n_{\text{cal}} = 35,737$, $n_{\text{test}} = 88,942$. Item-level tasks (i-plant, i-shippoint, i-incoterms): $n_{\text{val}} = 293,896$, $n_{\text{cal}} = 146,948$, $n_{\text{test}} = 402,855$. At these calibration set sizes, the finite-sample correction is negligible: $q - (1 - \alpha) < 3 \times 10^{-5}$ for all tasks, so finite-sample effects do not contribute to the observed coverage gaps.

A.3 SHAP COMPUTATION

Method: TreeExplainer [Lundberg et al., 2020]. Subsample: 10,000 per split. Aggregation: mean absolute SHAP across classes. Concentration metric (Eq. 2): computed on validation data only.

A.4 BASELINE DIAGNOSTICS

Native feature importance concentration uses a single model (seed=42) with LightGBM’s gain-based importance; concentration is top-1 gain divided by total gain. Ensemble disagreement is the standard deviation of validation coverage across 50 seeds. Both are computed without SHAP.

A.5 COMPUTATIONAL RESOURCES

Standard CPU (8 cores, 8GB RAM). Wall-clock: ~ 3 –4 hours for full 8-task, 50-seed suite. Software: Python 3.9, LightGBM 3.3, SHAP 0.41.

A.6 EXTERNAL DATASET PROTOCOLS

Table 6 retains the reported external dataset protocols. The supplied code does not establish a common pipeline: SALT and external coverage use different set conventions, validation labels are reused for tuning, and most external loaders/results are absent. In particular, the supplied KDD loader contradicts the previously stated official-test protocol, as noted below. Historical results need provenance reconciliation and a common-protocol rerun.

Table 6: External Dataset Protocols. Historical reported protocols; supplied external code halves validation for calibration/evaluation; n_{tr} , n_{te} = approximate train/test sizes used; Seeds = 10 for all external datasets. Type: **DS** = documented real-world shift; **NC** = null-shift control (random/pre-defined split, no genuine shift; low concentration and robust coverage are the expected and observed outcome).

Dataset	Shift mechanism	Type	K	n_{tr}	n_{te}	Source
Covertypes	Geographic: areas 1–2 (train), area 3 (val/cal), area 4 (test)	DS	7	290,680	36,968	[Dua and Graff, 2017]
Gas Sensor	Temporal batch drift: batches 1–6 (train), 7–8 (cal), 9–10 (test)	DS	6	9,840	4,070	[Dua and Graff, 2017]
KDDCup99	Supplied code: positional 60/20/20 of one 10% file; shift unverified	—	11	395,073	98,769	[Dua and Graff, 2017]
PAMAP2	Population shift: subject-based split (unseen subjects at test)	DS	12	1,674,379	268,493	[Dua and Graff, 2017]
Shuttle	Null-shift control: UCI Statlog; 60/20/20 random split	NC	7	46,400	11,600	[Dua and Graff, 2017]
Avila Bible	Null-shift control: copyist identification; 50/50 random split	NC	12	10,430	10,437	[Dua and Graff, 2017]
Pendigits	Null-shift control: UCI pre-defined split (digit recognition)	NC	10	7,494	3,498	[Dua and Graff, 2017]
Satimage	Null-shift control: satellite image; 60/20/20 random split	NC	6	5,144	1,286	[Dua and Graff, 2017]
Stack Overflow	Category shift (3-class subset); excluded from $n = 16$ historical selected endpoint	—	3	65,105	127,370	[Dua and Graff, 2017]

For Covertypes, the supplied loader uses wilderness areas 1–2 for training, 3 for validation and 4 for testing. These geographic indicators are not the cover-type labels (e.g., Spruce-Fir or Lodgepole Pine). For KDDCup99, the supplied loader fetches one 10% dataset, filters classes with at least 100 observations over the whole file, encodes object columns before splitting, and uses positional 60/20/20 partitions; no separate official test file is fetched. Historical table sizes and results need reconciliation with run provenance. The remaining reported splits are retained as historical descriptions where loaders are absent; SHAP uses the validation pool, which the external code also uses for early stopping before calibration/evaluation.

B PLACEBO TEST DETAILS

C HIGH MODEL VARIANCE ANALYSIS

Tasks marked with * in Table 2 exhibit extreme model variance ($CV > 50\%$):

- **s-group**: Mean 12.4%, median **0.5%**. Most models fail; outliers inflate mean.
- **i-shippoint**: Mean 72.7%, median **89.9%**. Most models succeed; failures deflate mean.

Table 7: Placebo Test: Pre-COVID vs COVID Degradation (50-Seed Mean Drops).

Task	Placebo	COVID	Ratio
s-shipcond	0.5%	71.6%	143×
s-group	2.0%	71.2%	36×
s-payterms	0.0%	77.1%	>100×
i-plant	1.8%	10.6%	6×
i-shippoint	1.5%	18.5%	12×
s-incoterms	1.2%	8.5%	7×
i-incoterms	0.6%	11.3%	19×
s-office	0.1%	0.1%	1×

D THRESHOLD SENSITIVITY ANALYSIS

Table 8 shows threshold performance across $n = 16$ multiclass tasks. Note that the 40% threshold is derived from the SALT concentration gap (in-sample); the 8 external datasets are fully out-of-sample. On the external subset alone at 40%: Precision = 1.00, Recall = 0.50 (1 TP: Covertypes; 1 FN: KDDCup99; 6 TN). Low external recall reflects sparse catastrophic cases; the primary claim rests on the $n = 16$ correlation ($\rho = 0.853$) rather than threshold classification. Ground-truth labels use the at-risk rule (mean coverage drop >15 pp, consistent with 50-seed means in Table 2). For high-variance tasks, the mean and median labels can differ: i-shippoint has mean drop 18.5 pp (At-risk) but median drop 1.2 pp (robust); Table 2 labels it ROB by median for descriptive purposes, while the framework uses the mean-based threshold for threshold evaluation.

Table 8: Threshold Sensitivity for SHAP Concentration Across 16 Multiclass Tasks (8 SALT + 8 external; Step 2 only, no protective-factor check). At-risk defined as coverage drop >15%.

Threshold	Prec	Rec	F1	TP	FP	FN
25%	0.63	0.83	0.71	5	3	1
30%	0.71	0.83	0.77	5	2	1
35%	0.83	0.83	0.83	5	1	1
40%	0.83	0.83	0.83	5	1	1
45%	1.00	0.83	0.91	5	0	1
50%	1.00	0.33	0.50	2	0	4

At 40%, the single FP is s-office (concentration 42.6%) which has a protective secondary feature (SALESORG Jaccard = 0.61), and the single FN is KDDCup99 (intermediate regime; $C = 21.1\%$, drop = 15.9 pp). At 30%, the two FPs are s-office and Shuttle ($C = 30.7\%$); at 25%, i-incoterms ($C = 28.9\%$) is additionally flagged. Applying the protective-factor check at 40% removes s-office (precision = 1.00) but leaves the KDDCup99 FN (recall = 0.83).

D.1 LEAVE-ONE-OUT CROSS-VALIDATION

To assess whether the 40% threshold generalizes beyond post-hoc selection, we perform leave-one-out cross-validation on the SALT dataset ($n = 8$). For each fold, we exclude one task, compute the optimal threshold from the remaining 7 tasks, and predict the held-out task’s vulnerability status.

Summary. LOO-CV achieves 87.5% accuracy (7/8 correct predictions). The threshold remains stable across folds: $43.1\% \pm 5.0\%$ (mean $\pm$ std). The effect size separating failed from robust tasks is Cohen’s $d = 3.08$ (large), with bootstrap 95% CI [0.31, 1.00] for the underlying correlation.

Effect size analysis. Failed tasks ($n = 4$) have mean concentration $50.2\% \pm 2.6\%$, while robust tasks ($n = 4$) have mean concentration $29.8\% \pm 7.7\%$. The separation of 20.4 pp yields Cohen’s $d = 3.08$ (large effect), describing separation in these retrospectively labeled eight tasks; it does not establish prospective reliability.

Misclassification analysis. Sales-office is a false positive under the concentration-only rule. Its stable-feature explanation is a hypothesis derived from this single case. The historical text inconsistently describes SALESORGANIZATION as dominant or secondary and its Jaccard overlap as train-test or train-validation; the original feature ledger is required to

Table 9: LOO-CV Per-Fold Predictions (SALT, $n = 8$). Threshold = optimal threshold from 7 training tasks; Pred = predicted status at that threshold; Actual = observed outcome (Failed if drop > 15 pp).

Task	Conc. (%)	Drop (pp)	Thresh (%)	Pred	Actual	Correct
s-shipcond	50.7	71.6	45	Vuln	Failed	✓
s-group	47.3	71.2	45	Vuln	Failed	✓
s-payterms	54.2	77.1	45	Vuln	Failed	✓
i-plant	23.9	10.6	45	Robust	Robust	✓
i-shippoint	48.8	18.5	45	Vuln	Failed	✓
s-incoterms	23.7	8.5	45	Robust	Robust	✓
i-incoterms	28.9	11.3	45	Robust	Robust	✓
s-office	42.6	0.1	30	Vuln	Robust	×

resolve this. A deployable protective-factor rule must use information available before test outcomes. High concentration is not necessary for failure, as the reported low-concentration KDD false negative demonstrates.

E DECISION FRAMEWORK VALIDATION

Table 10: Concentration Check Applied to All Multiclass Tasks (Actual = At-risk if drop > 15 pp).

Task	Conc. (%)	Step 2	Actual (> 15 pp)	Note
<i>SALT (in-sample)</i>				
s-payterms	54.2	VULN	At-risk	—
s-shipcond	50.7	VULN	At-risk	—
i-shippoint	48.8	VULN	At-risk*	High variance [†]
s-group	47.3	VULN	At-risk*	—
s-office	42.6	VULN	ROB	Protective [‡]
i-incoterms	28.9	ROB	ROB	—
i-plant	23.9	ROB	ROB	—
s-incoterms	23.7	ROB	ROB	—
<i>External (held-out)</i>				
Coverttype	49.8	VULN	At-risk	10/10 seeds
Shuttle	30.7	ROB	ROB	9/10 seeds
Avila Bible	20.5	ROB	ROB	10/10 seeds
PAMAP2	19.2	ROB	ROB	10/10 seeds
KDDCup99	21.1	ROB	At-risk [§]	Seed-dependent (FN)
Pendigits	14.5	ROB	ROB	10/10 seeds
Satimage	9.0	ROB	ROB	10/10 seeds
Gas Sensor	7.3	ROB	ROB	10/10 seeds
Stack Overflow	48.9	VULN	ROB	Excluded-scope false positive [¶]

[†]Median drop 1.2%; most seeds maintain coverage. [‡]SALESORGANIZATION: train-validation Jaccard=0.61, importance>15%.

[§]Concentration below threshold ($C = 21.1 \pm 7.5\%$), but mean drop exceeds the at-risk cutoff (15.9 pp); outcome seed-dependent (range −0.8 to 73.5 pp). In-sample F1 = 0.89 (1 FP: s-office). External: Coverttype deterministic catastrophic (10/10); five low-concentration datasets deterministic robust (10/10); Shuttle near-deterministic (9/10); KDDCup99 is the intermediate-regime false negative. [¶]Stack Overflow ($K = 3$, $C = 48.9\%$) is a reported false positive with drop −7.0 pp. Its exclusion from the historical $n = 16$ subset is not justified by a general APS ceiling: binary and three-class predictors can also lose all coverage under shift. The all-17 association is $\rho = 0.654$, $p = 0.004$.

Seed stability protocol. KDDCup99’s multi-seed analysis ($C = 21.1 \pm 7.5\%$) reveals that single-seed SHAP concentration can be unstable. We recommend computing C over $k \geq 5$ seeds and using the mean. Tasks in the ambiguous zone (35–45%) should receive additional scrutiny: if the upper 95% CI exceeds 40%, treat as potentially vulnerable. This safety override is designed to reduce KDDCup99-like false negatives in intermediate regimes. For SALT tasks, concentration is stable across 50 seeds (within-task CV < 5%); instability arises primarily when the top-feature identity varies across seeds.

Threshold selection. The 40% threshold is proposed based on the empirical distribution of SHAP concentration values on

validation data: a natural gap separates low-concentration tasks (24–29%) from high-concentration tasks (43–54%), visible in Table 10. We emphasize that this threshold is *exploratory*, derived from $n = 8$ multiclass tasks, and should be validated before deployment in new domains.

Cross-domain transfer test. The 40% threshold applied without tuning to 9 external domains yields 7/9 deterministic or near-deterministic outcomes (6 at 10/10, Shuttle at 9/10), with Covertypes correctly flagged as catastrophic and Gas Sensor correctly unflagged as robust. KDDCup99 is the principal intermediate-regime false negative, while Stack Overflow (3 classes) is a false positive excluded from the selected endpoint. Across the $n = 16$ multiclass set: $\rho = 0.853$, $p < 0.001$ (Section 6.5). The sensitivity analysis in Appendix D reports threshold performance across 30–50%; given sparse catastrophic external cases, this threshold should be treated as provisional.

F PRE-DEPLOYMENT VS POST-HOC BASELINES

SHAP concentration is computed on validation data *before* deployment. Prediction entropy and ECE require observing test data. This section reports entropy and ECE changes for completeness, noting that they are not available as pre-deployment diagnostics.

Table 11: Pre-Deployment (SHAP) vs Post-Hoc (Entropy, ECE) Shift Diagnostics (10-Seed ACI Experiments). SHAP concentration is computed on validation data only. Δ Entropy and Δ ECE require test-time observations.

Task	Cat	SHAP Conc. [†]	Δ Entropy	Δ ECE
<i>Severe tasks</i>				
s-shipcond	SEV	50.7%	−0.746	+0.408
s-group	SEV	47.3%	−1.274	+0.176
s-payterms	SEV	54.2%	−1.979	+0.629
<i>Robust tasks</i>				
i-plant	ROB	23.9%	+0.181	+0.131
i-shippoint	ROB	48.8%	+0.469	+0.073
s-incoterms	ROB	23.7%	+0.364	+0.350
s-office	ROB	42.6%	+0.032	+0.104

[†]Pre-deployment diagnostic (computed on validation data only). item-incoterms omitted (3-seed pilot only).

Counter-intuitively, all three catastrophic tasks (coverage drop $> 70\%$) show *decreasing* prediction entropy ($\Delta H < 0$): the model becomes *more* confident at test time, not less. This occurs because concentrated dependence on an OOD feature produces arbitrary but confident predictions. In contrast, the two robust tasks with non-trivial drops (i-plant, i-shippoint; coverage drops of 10–19%) show *increasing* entropy ($\Delta H > 0$), consistent with standard distributional broadening. A practitioner monitoring only entropy would correctly detect shift in moderate cases but would be *misled* in the most damaging catastrophic cases—observing decreasing uncertainty while coverage collapses. ECE increases across all tasks but does not discriminate between severity levels (both catastrophic and robust tasks show increases of +0.07 to +0.63). In this SALT comparison, SHAP concentration is the only evaluated pre-deployment metric that flags all three catastrophic tasks.

G TASK INDEPENDENCE ANALYSIS

We analyze pseudo-replication in the 50-seed design using intraclass correlation (ICC).

Between-task ICC. The archived summary reports $ICC = 0.675$ for coverage drops and 0.720 for test coverage. These describe a fitted variance decomposition across task/seed levels; they do not prove that tasks sharing one benchmark are independent. A design-effect conversion of 400 seed/task measurements cannot increase the number of independently observed tasks beyond the eight available tasks.

Per-task ICC(val,test). The previous analysis interpreted negative one-way agreement ICC as negative seedwise correlation and converted 50 pairs into effective sample sizes of 55–250. That inference is invalid. For example, when test coverage equals validation coverage minus a constant, Pearson correlation is one while the supplied agreement ICC can be strongly negative. We therefore withdraw the effective-sample-size table and the assertion that paired Wilcoxon tests must be conservative. Appropriate paired inference uses the distribution of seedwise differences and remains conditional on the common data.

Multiple comparisons. The reported concentration search includes eight metrics. Using the unrounded SALT p -value, the smallest Holm-adjusted value is $8 \times 0.0101755 = 0.0814$. The original five-metric family would give 0.0509, not a value below 0.05. Intermediate and pooled subset analyses are retained as exploratory; reusing discovery tasks and selecting endpoints cannot be repaired solely by correcting three subset p -values.

Multi-seed aggregation. The primary $n = 16$ correlation ($\rho = 0.853$) uses 10-seed means for all external datasets and 50-seed means for SALT tasks. The early $n = 11$ result ($\rho = 0.909$) used single-seed external values; switching to multi-seed means at $n = 11$ yields $\rho = 0.818$, $p = 0.002$, with the reduction driven by KDDCup99 whose multi-seed mean drop (15.9 pp) exceeds its single-seed value (-0.8 pp). This intermediate-concentration regime ($C = 21.1 \pm 7.5\%$) exhibits high outcome variance characteristic of the boundary between robust and vulnerable. Extending to $n = 16$ with consistent multi-seed means yields $\rho = 0.853$ ($p < 0.001$). Including Stack Overflow (3 classes, excluded scope subset) to make $n = 17$ weakens correlation to $\rho = 0.654$.

H THEORY DETAILS AND SCORE CDFS

Scope of the proof. The proof in Section 4 uses a fixed deterministic rank order and a probability-mixture weight λ . It does not assume that classes tied with the true label are strictly lower in probability, and it handles $\lambda = 1$ separately. Its coverage statement is for the explicit score-inversion set with fixed q , not a claim about recalibrated models or the archived crossing-label implementation.

Illustrative bound calculation, not empirical verification. The historical comparison substituted mixture weight $\lambda = C$, $\varepsilon = 0$ and $\bar{h} = 1/K$, obtaining numbers below five reported mean scores. None of these substitutions verifies the probability-mixture assumptions. In particular the lower bound decreases with $\bar{h}$, so $1/K$ is not conservative for every residual model whose true-class probability is at least uniform. These comparisons do not validate a theorem for measured TreeSHAP concentration; empirical verification requires an identified residual decomposition and a consistent score/set implementation.

I CONFORMITY SCORE STOCHASTIC DOMINANCE

Table 12 reports two-sample KS statistics comparing calibration and test APS conformity scores (seed=42, non-randomized APS). Higher conformity scores indicate the model requires more cumulative probability mass to reach the true label—i.e., the model is less confident about the correct class.

Table 12: KS Tests: Calibration vs. Test Conformity Score Distributions (Seed=42).

Task	KS	p	Mean _{cal}	Mean _{test}
<i>Catastrophic tasks (test scores > calibration)</i>				
s-shipcond	0.956	$< 10^{-10}$	0.54	0.98
s-payterms	0.748	$< 10^{-10}$	0.65	0.92
s-group	0.676	$< 10^{-10}$	0.98	1.00
<i>Robust tasks</i>				
i-plant	0.741	$< 10^{-10}$	0.73	0.86
s-incoterms	0.633	$< 10^{-10}$	0.75	0.87
s-office	0.994	$< 10^{-10}$	1.00	1.00
i-incoterms	0.559	$< 10^{-10}$	0.78	0.66
i-shippoint	0.505	$< 10^{-10}$	0.71	0.58

All KS tests are two-sided. Historical deterministic APS scores; a conservative split-conformal guarantee requires the held-out/exchangeability assumptions absent from the supplied protocol. For catastrophic tasks, test scores reportedly shift rightward; two-sided KS and mean shifts alone do not establish stochastic dominance or verify Theorem 1. Two robust tasks (i-incoterms, i-shippoint) show test scores shifted *leftward*—the model becomes more confident during COVID, possibly due to reduced product diversity.

J RAPS COMPARISON

Table 13 compares APS and RAPS coverage drops across all 8 tasks with 10-seed means ($\lambda = 0.01$, $k_{\text{reg}} = 5$).

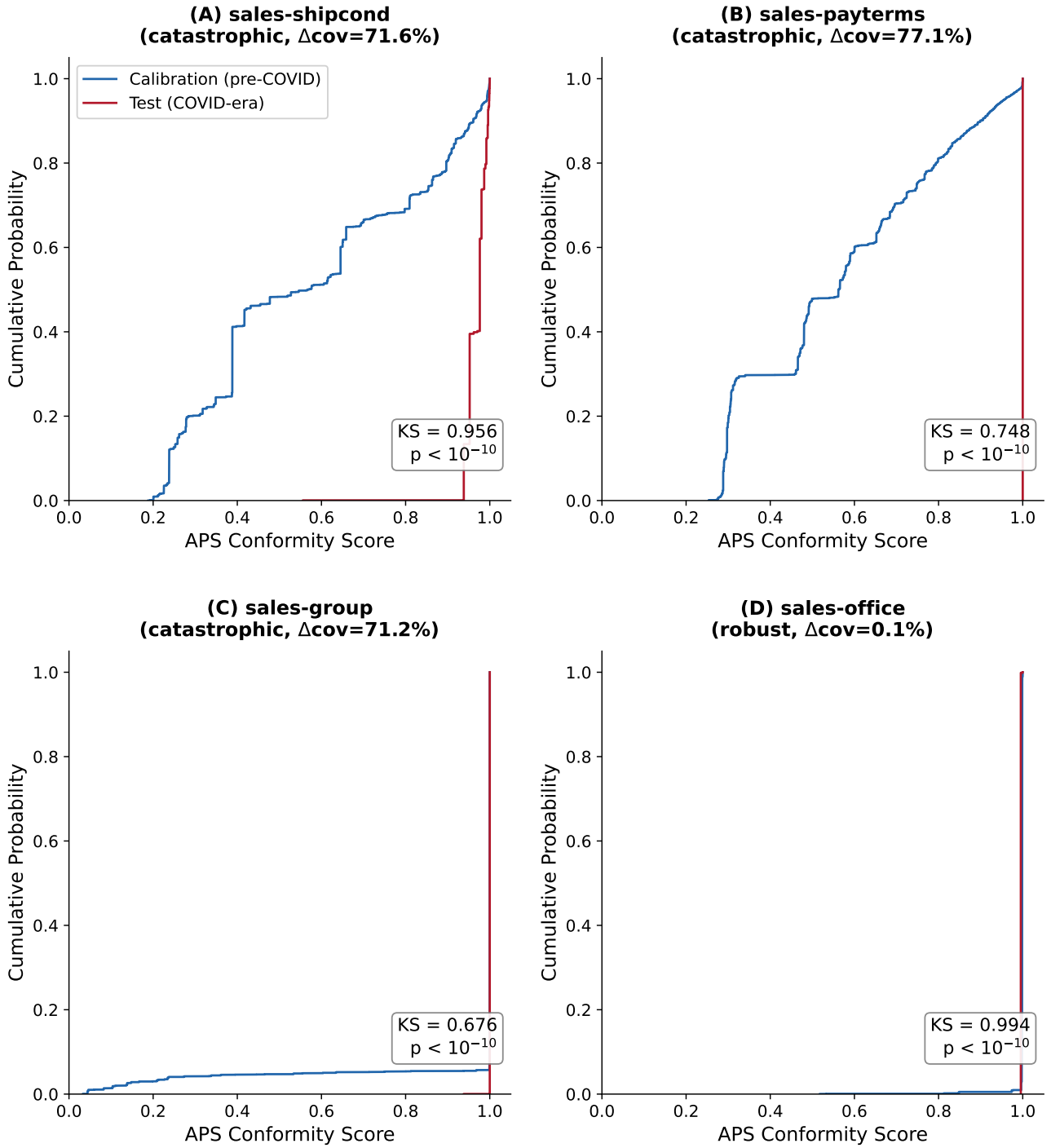


Figure 3: **Conformity score CDFs (seed=42).** Catastrophic tasks show rightward test-score shifts (stochastic dominance), while robust tasks can show near-degenerate or left-shift behavior.

Table 13: APS vs. RAPS Coverage Drop (10-Seed Means $\pm$ Std). $\lambda = 0.01$, $k_{\text{reg}} = 5$.

Task	CI	APS Drop	RAPS Drop	Δ	Note
<i>Severe tasks</i>					
s-shipcond	45	60.4 $\pm$ 31.8	67.8 $\pm$ 25.4	+7.4	No improvement
s-group	459	73.5 $\pm$ 33.9	10.4 $\pm$ 1.3	−63.1	RAPS protects
s-payterms	137	79.9 $\pm$ 29.4	35.1 $\pm$ 28.6	−44.8	Partial recovery
<i>Non-severe tasks</i>					
i-plant	35	11.8 $\pm$ 9.0	13.1 $\pm$ 5.8	+1.3	Near-identical
i-shippoint	69	9.3 $\pm$ 27.6	20.5 $\pm$ 21.6	+11.2	RAPS worse
i-incoterms	13	10.8 $\pm$ 9.1	10.8 $\pm$ 9.1	0.0	Identical
s-incoterms	13	6.7 $\pm$ 6.8	6.7 $\pm$ 6.8	0.0	Identical
s-office	25	0.04 $\pm$ 0.03	0.04 $\pm$ 0.03	0.0	Identical

All values are 10-seed means $\pm$ std. APS drops differ from 50-seed Table 2 due to smaller seed range (10 vs. 50) and high variance. The key comparison is APS vs. RAPS within the same seed set. Note: RAPS experiments used a random 50/50 calibration split rather than the deterministic first-half/second-half split used in the main APS experiments (Appendix A); this protocol difference does not affect the within-experiment APS vs. RAPS comparison but means RAPS coverage values are not directly comparable to Table 2. RAPS eliminates the catastrophic drop for sales-group by truncating the class accumulation that amplifies score inflation in many-class settings (mean set size: 232/459 classes). Sales-shipcond is unhelpt because single-feature dependence places dominant mass on the wrong class *before* the $k_{\text{reg}} = 5$ penalty activates—the exchangeability violation occurs at the top of the APS ordering, not through class accumulation. For low-class tasks (≤ 35 classes), RAPS and APS produce identical or near-identical drops, confirming RAPS regularization has negligible effect when prediction sets are already small. Item-shippoint ($K = 69$) is an exception: RAPS worsens coverage by 11.2 pp (paired Wilcoxon $p = 0.064$, 10 seeds), reflecting that the regularization penalty can disrupt calibration for intermediate-cardinality tasks with high seed variance.

K MODEL CLASS SENSITIVITY

To test whether SHAP concentration is diagnostic across model classes, we replicate the analysis with `RandomForestClassifier` (500 trees), `XGBClassifier`, and `CatBoostClassifier` (ordered boosting), all with `seed=42` and the same data pipeline (Table 14).

Table 14: SHAP Concentration and Coverage Drop Across Model Classes.

Task	LGB	RF	XGB	CB	LGB	RF	XGB	CB
	Concentration (%)				Coverage Drop (%)			
s-shipcond	50.7	57.4	44.6	53.7	71.6	15.8	47.2	1.1
s-group	47.3	22.5	33.3	31.8	71.2	50.0	56.8	14.5
s-payterms	54.2	27.9	30.6	25.7	77.1	13.8	20.4	−5.3
i-shippoint	48.8	52.1	41.0	43.1	18.5	22.7	80.4	61.8
i-plant	23.9	26.8	26.0	29.0	10.6	15.3	9.7	8.5
s-incoterms	23.7	26.8	21.6	28.2	8.5	2.5	3.7	−1.5
i-incoterms	28.9	17.5	18.4	21.7	11.3	2.9	1.1	−0.4
s-office	42.6	24.5	36.8	33.5	0.1	0.0	0.0	−0.0

Results. The diagnostic’s predictive power varies by model class (Table 15). All three gradient-boosted implementations show positive correlation: LightGBM ($\rho = 0.833$, $p = 0.010$), CatBoost ($\rho = 0.667$, $p = 0.071$), and XGBoost ($\rho = 0.548$, $p = 0.16$). RandomForest ($\rho = 0.30$, $p = 0.47$) does not replicate.

Mixed-effects analysis (historical, unverified). The archived summary fits task random intercepts to 24 task/model combinations from eight tasks and reports $\beta_1 = 1.64$, Wald 95% CI [0.70, 2.58], $p = 0.0006$. With eight clusters, the adequacy of the asymptotic interval/test is uncertain. A previously suggested Kenward–Roger p range was not computed and is withdrawn. The confidence interval and p -value share assumptions; one cannot claim the interval excludes zero under an uncomputed alternative correction. The original fitting code, model diagnostics and task/model data are missing. Random intercepts alone also do not establish independence of tasks sharing the benchmark.

Interpretation. The historical model/task summaries show heterogeneous outcomes and attribution profiles. They do not establish why RF or MLP correlations differ, nor that differences between correlations are statistically significant.

Table 15: Concentration–Drop Relationship by Model Class.

Model	Method	Conc.–Drop ρ	p
LightGBM	Boosting (GOSS)	0.833	0.010
CatBoost	Boosting (ordered)	0.667	0.071
XGBoost	Boosting (histogram)	0.548	0.16
RandomForest	Bagging	0.30	0.47
Mixed-effects, 3 boosting ($n = 24$)		$\beta_1 = 1.64$	<0.001
Mixed-effects, all 4 ($n = 32$)		$\beta_1 = 1.06$	0.010

Architectural explanations involving bagging smoothness or boosting reinforcement are hypotheses, requiring controlled comparisons and common attribution definitions.

The diagnostic is *model-specific*: it measures how a particular model’s learned dependence structure creates vulnerability. Practitioners should compute concentration for their *deployed* model class rather than transferring results across model families.

Neural network extension (historical, unverified). The reported two-layer MLP uses permutation importance rather than TreeSHAP, with 10 seeds for sales tasks and 3 for item tasks. Its concentration ratio is therefore not the same attribution estimand as the tree-model SHAP ratio. The reported $\rho = 0.43$, $p = 0.29$ cannot isolate architecture from attribution-method differences. High-drop low-concentration examples motivate a possible multi-feature mechanism but do not prove it; original code/results are required to assess the claim.

K.1 MLP VS LIGHTGBM TASK-LEVEL COMPARISON

Table 16 provides a structured comparison of SHAP concentration and coverage drop between MLP and LightGBM across all 8 SALT tasks.

Table 16: Task-Level Comparison: MLP vs LightGBM. MLP_C = MLP SHAP concentration; MLP_Drop = MLP coverage drop (pp); LGB_C = LightGBM SHAP concentration; LGB_Drop = LightGBM coverage drop (pp); Mode = failure mode classification.

Task	MLP_C	MLP_Drop	LGB_C	LGB_Drop	Mode
s-group	27.5%	78.4pp	47.3%	71.2pp	Global
s-payterms	28.0%	60.6pp	54.2%	77.1pp	Global
i-shippoint	77.2%	82.3pp	48.8%	18.5pp	Conc
s-incoterms	30.3%	42.3pp	23.7%	8.5pp	Mixed
s-shipcond	53.1%	67.1pp	50.7%	71.6pp	Conc
i-incoterms	31.0%	24.1pp	28.9%	11.3pp	Mixed
i-plant	28.2%	26.7pp	23.9%	10.6pp	Mixed
s-office	0.0%	0.1pp	42.6%	0.1pp	N/A

Mode: Conc = concentrated-dependence failure ($C > 40\%$ predicts drop); Global = global-sensitivity failure ($C \leq 40\%$ yet catastrophic drop); Mixed = intermediate regime; N/A = no meaningful drop in either model. Critical observation: s-group and s-payterms show MLP $C \approx 28\%$ yet 60–78pp drops, demonstrating the global-sensitivity failure mode where SHAP concentration does not capture the MLP’s vulnerability mechanism.

K.2 FAILURE MODE DEFINITIONS AND QUANTITATIVE CRITERIA

We formalize two distinct failure modes that determine when SHAP concentration is diagnostic.

Definition 1: Concentrated-Dependence Failure

Conditions: SHAP concentration $C > 40\%$; dominant feature undergoes distribution shift.

Diagnostic status: **Applies**—SHAP concentration correctly predicts vulnerability.

Characteristic: Model depends on a single feature whose OOD shift causes coverage collapse.

Definition 2: Global-Sensitivity Failure

Conditions: SHAP concentration $C \leq 40\%$; coverage degrades via coordinated multi-feature sensitivity.

Diagnostic status: **Does NOT apply**—SHAP concentration fails to predict vulnerability.

Characteristic: Model distributes importance but fails through global sensitivity to shift; requires alternative diagnostics (Table 19).

Table 17: Quantitative Criteria for Failure Mode Classification.

Criterion	Concentrated	Global
Top-1 SHAP	$C > 40\%$	$C \leq 40\%$
Gini coefficient	> 0.6	≤ 0.6
Top-3 cumulative	$> 70\%$	$\leq 70\%$

Concentrated failure mode: SHAP concentration diagnostic applies. Global failure mode: alternative diagnostics required (see Table 19).

Table 18: Decision Rule: When Does the SHAP Concentration Diagnostic Apply?

Model Class	Condition	Prediction	Recommendation
GBM/XGBoost/ LightGBM/CatBoost	$C > 40\%$	VULNERABLE	Mitigate or monitor
	$C \leq 40\%$	Lower risk	Standard deployment
MLP/Neural Net	$C > 40\%$	VULNERABLE (likely)	Mitigate or monitor
	$C \leq 40\%$	<i>Inconclusive</i>	Use gradient-based methods

The diagnostic is validated for gradient-boosted classifiers ($\rho = 0.833$, $p = 0.010$ for LightGBM). For MLPs, low concentration does not ensure robustness due to global-sensitivity failures ($\rho = 0.43$, $p = 0.29$).

K.3 ALTERNATIVE DIAGNOSTICS FOR MLPs WHEN $C \leq 40\%$

When SHAP concentration indicates low risk ($C \leq 40\%$) but the model is an MLP or other neural network, the following alternative diagnostics may capture global-sensitivity failures (Table 19).

Table 19: Alternative Diagnostics for Global-Sensitivity Failures in Neural Networks.

Diagnostic	Description
Gradient sensitivity	$S_j = \mathbb{E}[\ \partial f / \partial x_j\ _2]$: average gradient magnitude per feature
Feature ablation	$\Delta_j = \text{Cov}(D) - \text{Cov}(D x_j = \mu_j)$: coverage change when ablating feature j
Lipschitz bound	$\text{Cov}_{\text{test}} \geq \text{Cov}_{\text{val}} - L \cdot W_2(D_{\text{val}}, D_{\text{test}})$: coverage lower bound via Lipschitz constant L and Wasserstein-2 distance

These diagnostics target global-sensitivity failures where coordinated multi-feature shifts drive coverage degradation. They are proposed as complements to SHAP concentration for neural network deployments; empirical validation is left to future work.

L HYPERPARAMETER SENSITIVITY ANALYSIS

To assess whether the SHAP concentration–coverage drop correlation is robust to hyperparameter choices, we evaluate four LightGBM configurations on the 8 SALT multiclass tasks (Table 20).

Results. Across all four hyperparameter configurations, the concentration–drop correlation remains strong and significant: $\rho = 0.821 \pm 0.012$ (mean $\pm$ std), with $p \leq 0.015$ for all configurations. The optimal threshold varies minimally ($45.6\% \pm 1.1\%$), and all configurations achieve perfect F1 = 1.0 for at-risk classification on SALT tasks.

Interpretation. The four reported hyperparameter settings provide a small retrospective sensitivity summary. They do not establish a structural property of gradient boosting or robustness to unrestricted tuning. The generating artifacts are missing, and the ablation default uses 100 boosting rounds versus the main experiment’s stated maximum of 500; those protocols require reconciliation.

Table 20: Hyperparameter Sensitivity Analysis (SALT, $n = 8$ tasks). All configurations use boosting=gbdt, feature_fraction=0.8, bagging_fraction=0.8, bagging_freq=5.

Config	leaves	lr	n_est	ρ	p	Thresh	F1
Default	31	0.05	100	0.833	0.010	45.0%	1.0
Deeper	63	0.05	100	0.810	0.015	47.5%	1.0
Faster	31	0.10	100	0.810	0.015	45.0%	1.0
More	31	0.05	200	0.833	0.010	45.0%	1.0

leaves = num_leaves; lr = learning_rate; n_est = num_boost_round; Thresh = optimal threshold for at-risk classification (drop >15 pp); F1 = F1 score at that threshold.

L.1 PER-TASK CONCENTRATION STABILITY ACROSS HP CONFIGURATIONS

Table 21 shows per-task SHAP concentration values across all four hyperparameter configurations.

Table 21: Per-Task SHAP Concentration Stability Across HP Configurations. Std = standard deviation across four configurations.

Task	Default	Deeper	Faster	More	Std
sales-payterms	54.2%	57.3%	55.9%	54.3%	1.3%
sales-shipcond	50.7%	53.7%	52.3%	48.5%	1.9%
item-shippoint	48.8%	50.0%	50.7%	46.0%	1.8%
sales-group	47.3%	49.4%	48.5%	45.2%	1.6%
sales-office	42.6%	45.3%	43.0%	42.6%	1.1%
item-incoterms	28.9%	30.1%	28.9%	29.3%	0.5%
item-plant	23.9%	25.2%	24.4%	23.9%	0.5%
sales-incoterms	23.7%	25.4%	25.2%	22.5%	1.2%

Tasks sorted by Default concentration. Default = num_leaves=31, lr=0.05, n_est=100; Deeper = num_leaves=63; Faster = lr=0.10; More = n_est=200. Mean within-task std = 1.2%, demonstrating high stability of concentration values across HP configurations. The rank order of tasks by concentration is preserved across all four configurations.

M SHIFT CHARACTERIZATION

To characterize the nature of distribution shift in SALT tasks, we decompose shift into covariate and concept components using standard metrics (Table 22).

Table 22: Shift Characterization for SALT Tasks. Cov KS = mean Kolmogorov–Smirnov statistic across features (train vs. test); %Sig = fraction of features with KS $p < 0.05$; Label JS = Jensen–Shannon divergence of label distributions; Acc Drop = test accuracy minus validation accuracy.

Task	Cov KS	%Sig	Label JS	Acc Drop	Classification
s-shipcond	0.50	50%	0.083	+11.9%	Cov + Concept
s-group	0.50	50%	0.330	+1.0%	Cov (dominant)
s-payterms	0.50	50%	0.078	+4.0%	Cov (dominant)
i-plant	0.86	86%	0.144	−8.7%	Cov (dominant)
i-shippoint	0.86	86%	0.152	+13.1%	Cov + Concept
s-incoterms	0.50	50%	0.085	+7.6%	Cov + Concept
i-incoterms	0.86	86%	0.140	+63.4%	Cov + Concept
s-office	0.50	50%	0.026	−0.2%	Cov (dominant)

Cov = covariate shift; Concept = concept shift (change in $P(Y|X)$). Accuracy changes can be positive under shift when the test distribution favors majority classes.

Key findings.

- **Covariate shift is universal:** All SALT tasks exhibit substantial covariate shift ($\text{KS} \geq 0.50$, $\geq 50\%$ of features significantly different). This is expected given the COVID-19 temporal split.

- **Concept shift varies:** Label distribution shift (JS divergence) ranges from 0.026 (s-office) to 0.330 (s-group). Tasks with both high covariate shift and substantial label shift (s-shipcond, i-incoterms) tend toward catastrophic failure.
- **Accuracy is not diagnostic:** Accuracy changes are uninformative for coverage prediction—s-shipcond shows +11.9% accuracy improvement yet 71.6 pp coverage drop, while s-office shows −0.2% accuracy change and 0.1 pp coverage drop. This occurs because conformal prediction set size is sensitive to calibration–test score distribution mismatch, which accuracy does not capture.

Interpretation. Standard shift detection identifies covariate shift uniformly across all tasks (consistent with Section 6.3), while concept shift magnitude partially distinguishes catastrophic from robust outcomes. However, neither metric alone predicts failure severity as effectively as SHAP concentration ($\rho = 0.833$), which captures how the model’s learned dependence structure interacts with the specific features that shift. The worst outcomes occur when concentrated dependence targets a feature experiencing both covariate and concept shift.

M.1 PER-FEATURE COVARIATE SHIFT DETAILS

Tables 23 and 24 show per-feature KS statistics for sales-level and item-level tasks respectively.

Table 23: Per-Feature Covariate Shift for Sales-Level Tasks (6 features). KS = Kolmogorov–Smirnov statistic (train vs. test).

Feature	Shifted	KS
SALESDOCUMENTTYPE	No	0.0
SALESORGANIZATION	Yes	1.0
DISTRIBUTIONCHANNEL	Yes	1.0
ORGANIZATIONDIVISION	Yes	1.0
BILLINGCOMPANYCODE	No	0.0
TRANSACTIONCURRENCY	No	0.0

KS = 1.0 indicates complete distribution non-overlap between train and test. 3/6 features (50%) show significant covariate shift, consistent with the %Sig values in Table 22.

Table 24: Per-Feature Covariate Shift for Item-Level Tasks (7 features). KS = Kolmogorov–Smirnov statistic (train vs. test).

Feature	Shifted	KS
SALESDOCUMENTITEM	Yes	1.0
SALESDOCUMENTITEMCATEGORY	No	0.0
PRODUCT	Yes	1.0
SOLDTOPARTY	Yes	1.0
SHIPTOPARTY	Yes	1.0
BILLTOPARTY	Yes	1.0
PAYERPARTY	Yes	1.0

6/7 features (86%) show significant covariate shift, consistent with the %Sig values in Table 22. The high proportion of shifted features at item-level reflects COVID-19’s impact on supply chain transactions: new products, parties, and item identifiers entered the system during the pandemic period.

N WILDS BENCHMARK VALIDATION

To further validate the specificity of the SHAP concentration diagnostic, we evaluate on four additional benchmark datasets from the WILDS benchmark [Koh et al., 2021] and standard repositories. The historical summaries report low concentration and robust coverage in four evaluations. Their generating code and outputs are absent; Covertypes is reused under another split, and CivilComments/Adult are binary tasks outside the selected primary subset.

Dataset details.

- **CivilComments** (WILDS): Toxicity classification with demographic identity-based shift. TF-IDF features (1,000 dimensions). Low concentration (26.2%) correctly predicts maintained coverage.
- **Amazon Reviews** (WILDS): Product rating prediction with user/time domain shift. TF-IDF features (1,000 dimensions). Very low concentration (1.9%) correctly predicts robustness.

Table 25: WILDS and Standard Benchmark Validation (Specificity Test). Two evaluations use WILDS; two use standard datasets. Covertypes also appears in the primary analysis under a different split. All datasets exhibit low SHAP concentration (**all** <40%; mean = 17.0%) and robust coverage, reporting four true negatives among this small selected sample; independent specificity is not established. Shift types: Demographic = identity-based subgroup shift; User/Time = user and temporal domain shift; Geographic = regional covariates; Age = demographic age-based shift. Note: This table tests specificity (low- $C \rightarrow$ robust); for the catastrophic Covertypes case (high- $C \rightarrow$ failure), see Table 10.

Dataset	Source	Shift Type	Conc. (%)	Drop (pp)	Predicted	Actual
CivilComments	WILDS	Demographic	26.2	-0.7	Robust	Robust
Amazon	WILDS	User/Time	1.9	+0.4	Robust	Robust
Covertypes [†]	sklearn	Geographic	12.5	+8.0	Robust	Robust
Adult	OpenML	Age-based	27.5	+1.4	Robust	Robust

[†]Covertypes here uses a different split protocol from the main external validation (Table 6), which explains the different concentration and drop values. The main-text Covertypes uses wilderness areas 1–2/3/4 geographic split (concentration 49.8%, drop 81.8 pp); this validation uses a wilderness area 1 vs. others geographic split (concentration 12.5%, drop 8.0 pp). Both results are consistent with the diagnostic: high concentration predicts failure, low concentration predicts robustness.

- **Covertypes** (sklearn): Forest cover classification with geographic shift (wilderness area 1 vs. others). 54 cartographic features. Low concentration (12.5%) correctly predicts mild degradation.
- **Adult Income** (OpenML): Income prediction with age-based demographic shift (train: age<50, test: age≥50). 14 demographic/employment features. Low concentration (27.5%) correctly predicts maintained coverage.

Specificity validation. The four reported benchmark evaluations have low SHAP concentration—all below the 40% threshold (individual values: 26.2%, 1.9%, 12.5%, 27.5%; **mean concentration = 17.0%**). The diagnostic correctly predicts robustness for all four, giving an observed 4/4 true-negative rate, with substantial sampling and selection uncertainty and without raw-run verification. This complements the sensitivity evidence from SALT, where high-concentration tasks correctly predict catastrophic failure.

Combined validation accuracy (SALT + WILDS). Combining the 8 SALT tasks with these 4 WILDS benchmarks yields $n = 12$ total tasks (distinct from the $n = 16$ primary endpoint which uses 8 different external UCI datasets). The threshold accuracy is **91.7% (11/12)**:

- **Sensitivity** (SALT): 4/4 at-risk SALT tasks flagged under the >15 pp rule (historical descriptive counts)
- **Specificity** (WILDS + robust SALT): 7/8 robust tasks correctly unflagged (87.5%)
- **Single misclassification**: sales-office (concentration 42.6%, protective secondary feature)

These small, selected historical counts do not validate an operational boundary. Missing runs, binary tasks outside the primary scope and reuse of Covertypes require explicit treatment in a prospective evaluation.